Finite-resource cooling of qudits beyond thermodynamic length

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The heat dissipated when a quantum system is cooled with N thermal machines exceeds Landauer's bound; for swap-based collision models the excess is set by thermodynamic length. We show that for qudits the leading finite-resource cost is instead set by a Finsler geometry generated by resonant permutations of energy eigenstates. It improves on thermodynamic length near infinite temperature for every spectrum, and near the ground state by a factor approaching $d-1$. To leading order coherence does not help, and at infinite temperature d_M-level machines gain at most a factor d_M-1.

Landauer's principle bounds the heat dissipated when a quantum system is cooled or reset [1, 2], and the third law implies that the bound is reached only with diverging resources [3–5]. How the excess cost depends on the available resources has been studied in many settings, from finite-time versions of Landauer's bound [6–8] to heat-bath algorithmic cooling [9–12] and the limits of dynamic and autonomous refrigeration [13–15]. For slowly driven processes the excess is governed by thermodynamic length, the length of the protocol's path in a Riemannian metric on equilibrium states [16–21]. Taranto *et al.* [22] carried this picture over to Markovian collision models [23, 24], in which the system interacts in turn with N fresh thermal machines. For swaps with suitably chosen machine Hamiltonians, the excess dissipation is $(L^*)^2/2N + O(N^{-2})$, where L^* is the thermodynamic length between the initial and final Gibbs states. They proved this optimal over all unitaries for a qubit cooled by qubit machines, and their numerics for qutrits and ququarts suggested that it might hold in general [22].

Here we show that for qudits the finite-resource cost is set by a different geometry. The difference comes from resonances: when several machine transitions are resonant with the same system transition, or system transitions share a frequency, as in any equally spaced spectrum, a single collision can move population through several system–machine channels at once. Since the dissipation is quadratic in the population flow, dividing the flow lowers it, much as a current shared by two identical resistors dissipates half the heat, whereas a swap uses a single channel for each pair of levels. We prove that, to leading order around dissipationless collisions, coherent collisions are dominated by mixtures of resonant permutations of energy eigenstates (Theorem 1). The local cost is therefore set by a classical Finsler geometry generated by these permutations, which reduces to thermodynamic length for qubits. It improves on thermodynamic length near infinite temperature for every spectrum and near the ground state by a factor approaching $d-1$, and explicit protocols save up to 50% for qutrits and 83% for eight-level systems. At infinite temperature, d_M-level machines gain at most a factor d_M-1, which transfers between two levels attain (Theorem 2). Part of the gain, near the ground state and for transfers between the extreme levels, is the advantage of larger machines noted in Refs. [14, 22]; here it appears with machines of the same dimension as the system, and at high temperature it comes from commensurability alone.

Setting.—We follow Ref. [22]. A system S with Hamiltonian H_S starts in the Gibbs state $\tau_S(\beta)$ and is to be brought to $\tau_S(\lambda\beta)$, $\lambda>1$. In step $n=1,\dots,N$ it interacts through a unitary U_n with a fresh machine M_n, which starts in the Gibbs state $\tau_{M_n}(\beta)$ of a Hamiltonian H_{M_n} that we are free to choose. Unless stated otherwise the machines have the same dimension d as the system; Theorems 1 and 2 hold for any machine dimension d_M. The cost is the energy ΔE_M deposited in the machines. Writing ρ_n for the state of the system after step n, the equality form of Landauer's principle [2, 25, 26] gives

$$\beta\Delta E_M = \big[S(\rho_0)-S(\rho_N)\big] + \sum_{n=1}^{N}\Sigma_n, \qquad (1)$$
$$\Sigma_n = I(S{:}M_n)_{\rho'} + D(\rho'_{M_n}\,\|\,\tau_{M_n}),$$

where primes denote states after the collision. The entropy difference is fixed by the endpoints, so finite resources enter only through the dissipations $\Sigma_n \geq 0$. A swap with a machine prepared in the next state of the system, $\tau_{M_n}=\rho_n$, dissipates $\Sigma_n = D(\rho_{n-1}\,\|\,\rho_n)$, and the best swap protocol has total dissipation $\Sigma_N^{\mathrm{swap}} = (L^*)^2/2N + O(N^{-2})$ with $L^* = 2\arccos \mathrm{tr}\,\sqrt{\tau_S(\beta)}\sqrt{\tau_S(\lambda\beta)}$ [22].

Dissipation in one collision.—Each Σ_n depends only on the incoming state $\rho=\rho_{n-1}$ of the system, the machine state $\tau=\tau_{M_n}$ and the unitary $U=U_n$. Since $S(\rho'_{SM})=S(\rho\otimes\tau)$, it can be written as

$$\Sigma = D(\rho'_{SM}\,\|\,\rho\otimes\tau) - D(\rho'_S\,\|\,\rho), \quad \rho'_{SM}=U(\rho\otimes\tau)U^\dagger, \qquad (2)$$

because both sides equal $S(\rho'_S)-S(\rho\otimes\tau)-\mathrm{tr}\,\rho'_M\log\tau$ [Supplemental Material (SM), Sec. S1]. For a swap with $\tau=\rho'$ the first term is $D(\rho'\,\|\,\rho)+D(\rho\,\|\,\rho')$, and Eq. (2) gives back $D(\rho\,\|\,\rho')$.

Collision geometry.—We first restrict to permutation unitaries; Theorem 1 below shows that, to leading order, nothing is lost by doing so. Let $\rho = \mathrm{diag}(p)$ and $\tau = \mathrm{diag}(q)$, and let U permute the product states $|x, y\rangle$ (system level x, machine level y) according to a permutation π. The configuration weights $r = p \otimes q$ are then permuted, and Eq. (2) becomes $\Sigma = D(r \circ \pi^{-1} \,\|\, r) - D(p' \,\|\, p)$. The system changes little if π exchanges configurations of nearly equal weight, and we call π *resonant* at p if there is a machine state q^* with $r(c) = r(\pi(c))$ for every configuration c, so that nothing moves. Perturbing the machine state to $q = q^* + \delta q$ then moves the system by $\delta p = A_\pi\, \delta q$, at a cost

$$\Sigma = \tfrac{1}{2}\Big[\sum_c \frac{(\delta r_c)^2}{r_c} - \sum_x \frac{(\delta p_x)^2}{p_x}\Big] + O(\delta q^3), \qquad (3)$$

which is a quadratic form in δq (SM, Sec. S2). For the swap, $q^* = p$ and Eq. (3) reduces to $\frac{1}{2}\|\delta p\|_F^2$, where $\|\delta p\|_F^2 = \sum_x (\delta p_x)^2/p_x$ is the Fisher metric; its geodesic distance between Gibbs states is L^*. If a protocol uses permutations that remain resonant along a slow path $p(t)$, its cost is, to leading order, $\frac{1}{2N}(\int \|\dot p\|_{\mathrm{eff}}\, dt)^2$. Here $\|\cdot\|_{\mathrm{eff}}$ is the norm whose unit ball is the closed convex hull of the unit sets $\{\delta p : \Sigma \le \frac{1}{2}$ in Eq. (3)$\}$ of all pairs (π, q^*) resonant at p. In general it is a Finsler norm rather than a Riemannian one, and it can change discontinuously across resonance manifolds. Swaps, and the permutation π_{ext} introduced below, are resonant at every p, and thermodynamic length corresponds to using swaps only.

For qubits, only two resonant permutations move the system, the swap $|0,1\rangle \leftrightarrow |1,0\rangle$ and the exchange $|0,0\rangle \leftrightarrow |1,1\rangle$, and both give the Fisher norm (SM, Sec. S3), in agreement with Ref. [22]. Of the 9! qutrit permutations, 575 are resonant at a generic p. For the equally spaced qutrit cooled between $\beta\omega = 1$ and 3, the hull at interior points of the Fisher–Rao geodesic coincides with the Fisher ellipse in the direction of motion, so swaps are locally optimal along that path, consistent with the numerics of Ref. [22]. On the resonance manifold $p_1^2 = p_0 p_2$, however, which contains every Gibbs state of the equally spaced qutrit $H_S = \omega\, \mathrm{diag}(0, 1, 2)$, 234 further permutations become resonant, all of which move the system, and the hull extends beyond the Fisher ellipse.

Coherent collisions.—We now turn to arbitrary collisions, which may create coherence between energy eigenstates; in other settings coherence is known to contribute to entropy production and to raise the heat cost of erasure [7, 27]. A collision is dissipationless if and only if $U = (V \otimes 1)U_0$, with V a unitary on S and U_0 commuting with $\rho \otimes \tau^*$ for some machine state τ^* (SM, Sec. S10). Since V affects neither Σ nor the spectrum, we set $V = 1$. Now take a general collision close to such a one, $\tau = \tau^* + \epsilon\Theta$ and $U = U_0 e^{i\epsilon K}$, where Θ is diagonal (a rotation of the machine basis can be absorbed into U) and K is any Hermitian generator. Expanding Eq. (2) gives $\Sigma = \epsilon^2 Q + O(\epsilon^3)$ with

$$\begin{aligned} Q &= \tfrac{1}{2}\|X\|^2_{\rho\otimes\tau^*} - \tfrac{1}{2}\|\,\mathrm{tr}_M\, X\|^2_\rho, \\ X &= U_0 D U_0^\dagger - D + U_0\, i[K, \rho\otimes\tau^*]\, U_0^\dagger, \end{aligned} \qquad (4)$$

where $D = \rho \otimes \Theta$ and $\|\cdot\|_\sigma$ is the Kubo–Mori norm, the Hessian of the relative entropy [28].

Theorem 1. There exist permutations P_k of a product eigenbasis of $\rho \otimes \tau^*$ that commute with $\rho \otimes \tau^*$, i.e., are resonant, and weights $\lambda_k \ge 0$ with $\sum_k \lambda_k = 1$, such that to first order in ϵ the collision changes the spectrum of ρ by $\sum_k \lambda_k \delta_k$, and $Q \ge \sum_k \lambda_k Q_k$. Here δ_k and Q_k are the change of the spectrum and the cost of P_k with the same machine perturbation Θ.

The proof (SM, Sec. S10) splits X into a part inside the degenerate eigenspaces of $\rho \otimes \tau^*$, generated by Θ, and a part between different eigenspaces, generated by K. By monotonicity of the Kubo–Mori metric under the partial trace, the coherences created by the second part cost more in the first term of Eq. (2) than they give back in the second. For the first part, after a rotation inside the degenerate eigenspaces of ρ, only the doubly stochastic matrix $B_{ij} = |(U_0)_{ij}|^2$ matters. Birkhoff's theorem [29] writes it as $\sum_k \lambda_k P_k$, and the bound follows because the second term, $\frac{1}{2}\|\delta p\|_F^2$, is convex in δp. Since $Q_k \ge \frac{1}{2}\|\delta_k\|^2_{\mathrm{eff}}$ by definition of the hull, every collision close to a dissipationless one satisfies $\Sigma \ge \frac{1}{2}\|\delta p\|^2_{\mathrm{eff}} + O(\epsilon^3)$.

For complete protocols we consider the quasistatic regime, in which every collision is an $O(1/N)$ perturbation of one that is dissipationless for its incoming state, and all populations stay bounded away from zero. Summing over steps, any such protocol dissipates at least $\ell^2/2N$ to leading order, where $\ell = \sum_n \|\delta p_n\|_{\mathrm{eff}}$ is the length of its spectral path in the collision geometry. Alternating between resonant permutations attains this bound wherever they stay resonant along the path, as swaps, π_{A} and π_{ext} below do. For qubits this recovers, to leading order, the optimality of swaps among all unitaries [22]. Collisions applied $O(1/N)$ away from a resonance manifold are also dominated by permutations, but the offset could enlarge the hull; in an optimistic relaxation at six temperatures along the curve of Fig. 1(a), it does so by at most 3.6% (SM, Sec. S10).

Qutrits.—The mechanism is most transparent at infinite temperature, where all nine configurations have weight 1/9 and every permutation is resonant. The Gibbs curve leaves the maximally mixed state in the direction $\delta p \propto (1, 0, -1)$, moving population from level 2 to level 0. A swap achieves a transfer M at cost $3M^2$. The permutation

$$\pi_{\mathrm{A}}: \quad |0,0\rangle \leftrightarrow |2,2\rangle, \qquad |0,1\rangle \leftrightarrow |2,0\rangle \qquad (5)$$

instead routes the transfer through two channels that use different machine transitions. A machine perturbation

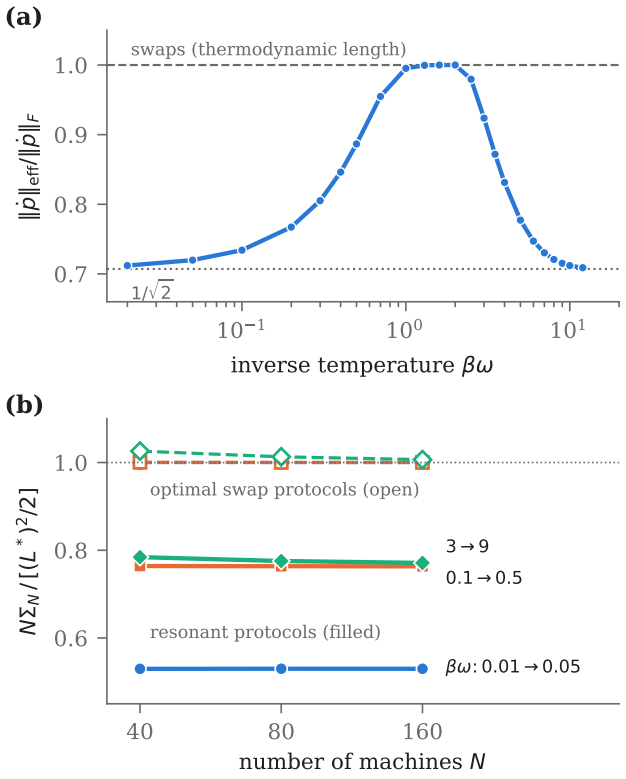


FIG. 1. (a) Cost per unit thermodynamic length, $\|\dot{p}\|_{\rm eff}/\|\dot{p}\|_F$, along the Gibbs curve of the equally spaced qutrit. Here $\|\cdot\|_{\rm eff}$ is the norm whose unit ball is the convex hull of the unit sets of all resonant permutations, computed from primal and dual cone programs that agree to within 2×10^{-7}. By Theorem 1 it is optimal among collisions close to dissipationless ones. Swaps give 1. The ratio tends to $1/\sqrt{2}$, i.e., half the dissipation, at high and at low temperature. (b) Exact finite-N protocols: dissipation $N\Sigma_N$ in units of the thermodynamic-length value $(L^*)^2/2$ for three cooling windows. Open symbols, optimal swap protocols (a convex program); filled symbols, resonant permutation protocols (SM, Sec. S5).

$\delta q = \epsilon(0, 1, -1)$ sends $m = \epsilon/3$ through each, and Eq. (3) gives $\Sigma = 9(m_1^2 + m_2^2) - 3(m_1 + m_2)^2 = \frac{3}{2}M^2$ for $m_1 = m_2 = M/2$, half the swap cost. No single permutation does better (SM, Sec. S4). For qutrit machines, a dual certificate over all 9! permutations and machine states shows that mixtures and alternations do not help either, so by Theorem 1 half the swap cost is optimal at infinite temperature among all qutrit-machine collisions close to dissipationless ones (SM, Sec. S10); Theorem 2 below extends this bound to every direction.

Away from infinite temperature, $\pi_{\rm A}$ stays resonant for every p, with machine populations $q \propto (1, e^{-2\beta\omega}, e^{2\beta\omega})$, i.e., an equally spaced machine with gap 2ω. At intermediate and low temperatures, however, the advantage comes from permutations that are resonant only on the manifold $p_1^2 = p_0 p_2$ (SM, Sec. S5). As $\beta\omega \to \infty$, a pair of 3-cycles, $\pi_{\rm C}$, routes the transfer between the two lowest levels through two degenerate excited levels of the machine, and an even split again halves the cost (SM, Sec. S6). The resulting ratio, shown in Fig. 1(a), is below 1 except for $1.3 \lesssim \beta\omega \lesssim 2$.

TABLE I. Exact protocols with 41 machines for equally spaced d-level systems. "Swap" is the optimum over swap protocols with the same number of machines and exact endpoints, a convex problem (SM, Sec. S5). "Resonant" uses the permutations listed in SM, Secs. S5, S7 and S9, and ends exactly in $\tau_S(\lambda\beta)$; it was evaluated in 40-digit arithmetic with explicit machine Hamiltonians $H_{M_n} = -\beta^{-1}\log q_n$. Dissipations are in units of 10^{-6}.

d	$\beta\omega \to \lambda\beta\omega$	swap	resonant	reduction
3	$0.01 \to 0.05$	13.001	6.911	46.8%
3	$0.1 \to 0.5$	1235.9	967.5	21.7%
3	$3 \to 9$	2288.5	1794.1	21.6%
3	$8 \to 16$	16.288	8.193	49.7%
4	$0.05 \to 0.25$	596.37	451.72	24.3%
4	$3 \to 6$	1533.6	872.2	43.1%
4	$8 \to 16$	16.288	6.060	62.8%
5	$8 \to 16$	16.288	4.613	71.7%
6	$8 \to 16$	16.288	3.730	77.1%
7	$8 \to 16$	16.288	3.137	80.7%
8	$8 \to 16$	16.288	2.714	83.3%

Exact protocols.—We also constructed complete protocols with exact dynamics. For each cooling window we fixed a pattern of permutations and optimized all machine populations, with the final state constrained to $\tau_S(\lambda\beta)$. A small residual ($\lesssim 10^{-7}$) is removed by one extra swap, which is included in the machine count. Each protocol was then re-evaluated in 40-digit arithmetic with explicit Hamiltonians, $d^2 \times d^2$ permutation unitaries and the energy balance (1). As a benchmark we minimized $\sum_n D(p_{n-1} \,\|\, p_n)$ over swap protocols; this function is jointly convex, so the benchmark is the global optimum over all machine Hamiltonians. In all cases of Table I the resonant protocol dissipates 22–83% less than the best swap protocol with the same number of machines, and the advantage persists at leading order in $1/N$ [Fig. 1(b) and SM, Sec. S5]. The resonant machines use level spacings up to about twice those of the swap protocols, but since the benchmark allows any machine Hamiltonian, the gain is not a matter of larger gaps.

Higher dimensions.—For the equally spaced ququart, permutations that conserve $H_S + H_M$ for integer machine spectra give cost ratios [as in Fig. 1(a)] between 0.72 and 0.97 at all five temperatures we tested (SM, Sec. S7), and exact protocols save 24% and 43% (Table I). For an equally spaced d-level system of any dimension, the permutation

$$\pi_{\rm ext}: \quad |0, y\rangle \leftrightarrow |d-1, y+1\rangle, \qquad y = 0, \ldots, d-2, \quad (6)$$

conserves $H_S + H_M$ for a machine with gap $(d-1)\omega$ and

is resonant at every p. At infinite temperature it moves population between the two extreme levels through $d-1$ parallel channels, at $1/(d-1)$ of the swap cost (SM, Sec. S8). The cooling direction $g_x \propto \bar{E}-E_x$ is not parallel to this transfer, but alternating π_{ext} with swaps realizes any convex combination of their unit sets. This beats swaps whenever the Fisher cosine between the cooling direction and the extreme transfer exceeds $(d-1)^{-1/2}$, which holds for all $d \geq 3$ (SM, Sec. S8). Near infinite temperature the leading coefficient is therefore at most 0.50, 0.65, 0.70 and 0.73 times the thermodynamic-length value for $d = 3$, 4, 5 and 6, and with 41 machines, exact protocols alternating swaps and π_{ext} save 33%, 28% and 23% for $d = 4$, 5 and 6 when cooling from $\beta\omega = 0.01$ to 0.05.

Near infinite temperature, equal spacing is unnecessary: for any spectrum $E_0 < \cdots < E_{d-1}$, π_{ext} is resonant at every p with a machine of level spacing $E_{d-1}-E_0$, and its cost ratio at infinite temperature is again $1/(d-1)$. Since $\sum_x (E_x-\bar{E})^2 \leq d(E_{d-1}-E_0)^2/4$, its Fisher cosine with the cooling direction is at least $\sqrt{2/d}$, which exceeds $(d-1)^{-1/2}$ for every $d \geq 3$ (SM, Sec. S8). Swaps are therefore suboptimal near infinite temperature for every spectrum, and for qutrits with unequal spacings, exact protocols cooling from $\beta = 0.01$ to 0.05 save 16–31%.

Toward the ground state.—Close to zero temperature, and for a non-degenerate first excited level, the cost is dominated by the transfer between the two lowest levels. Consider a machine with a ground level g and $d-1$ degenerate excited levels e_j at the lowest gap of the system, and the d-cycle $|0,e_0\rangle \to |1,g\rangle \to |0,e_1\rangle \to \cdots \to |0,e_{d-2}\rangle \to |0,e_0\rangle$. Its resonance condition only fixes $q_e/q_g = p_1/p_0$, so it is resonant at every p and for every spectrum. To leading order, the cost associated with $|1,g\rangle$ is canceled by the term $-D(\rho'_S\|\rho)$ of Eq. (2), while the outflow is shared among the $d-1$ configurations $|0,e_j\rangle$. The local cost of the dominant transfer is therefore $1/(d-1)$ of the swap value (SM, Sec. S9; for $d = 3$ the cycle is part of π_{C}). Exact protocols cooling from $\beta\omega = 8$ to 16 save 63–83% for $d = 4,\ldots,8$, close to the predicted saving $1-1/(d-1)$ (Table I and Fig. 2). The construction also works for unequal spacings, saving 46% for $E = (0,1,2.3)$ and 59% for $E = (0,1,1.7,3.1)$.

The asymptotic optimality of the thermodynamic-length protocol suggested by the numerics of Ref. [22] therefore fails for every $d \geq 3$ and every spectrum, even with machines of the same dimension as the system; Theorem 1 of Ref. [22], which concerns swap protocols, is unaffected. At intermediate temperatures the gain does rely on nearly commensurate levels: for $H_S = \omega\,\mathrm{diag}(0,1,2+\delta)$ and $\beta\omega = 0.1 \to 0.5$, the pattern $\pi_{\text{B}},\pi_{\text{A}}$ of SM Sec. S5 saves 23% at $\delta = 0.02$ and 15% at $\delta = 0.1$ but nothing at $\delta = 0.3$, where π_{ext} still saves 31% closer to infinite temperature.

Maximal gain.—Our second theorem limits the possible gain.

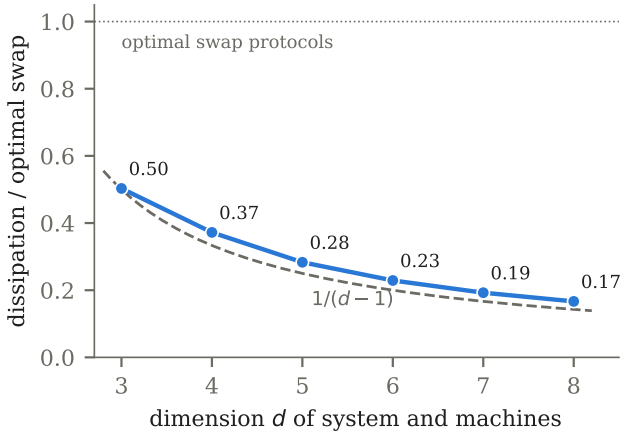


FIG. 2. Cooling toward the ground state ($\beta\omega = 8 \to 16$, 41 machines). Dissipation of the resonant protocol divided by that of the optimal swap protocol, for equally spaced d-level systems and d-level machines (filled symbols). Dashed line: $1/(d-1)$, the cold-limit value of the local analysis (SM, Sec. S9). The same value is the largest possible gain at infinite temperature (Theorem 2), and we conjecture that it bounds the gain at every temperature.

Theorem 2. Let the machines have d_M levels. At the maximally mixed state of the system, every collision close to a dissipationless one satisfies

$$\Sigma \;\geq\; \frac{1}{d_M-1}\,\tfrac{1}{2}\|\delta p\|_F^2 + O(\epsilon^3) \qquad (7)$$

for every δp. For qubit machines ($d_M = 2$), Eq. (7) holds at every state of the system and for every d.

By Theorem 1 it suffices to consider resonant permutations. For each system level x, the machine levels from which population flows into the configurations $|x,y\rangle$ define a map from the set of machine levels to itself. The flows cancel around every cycle of this map, so at least one machine level drops out of the transfer, and the Cauchy–Schwarz inequality gives the factor $(d_M-1)/d_M$ (SM, Sec. S11). Transfers between two levels reach the bound: with a uniform machine, the exchange $|0,y\rangle \leftrightarrow |d-1,y+1\rangle$, $y = 0,\ldots,d_M-2$, attains it for every d and d_M (for $d_M = d$ this is π_{ext}), and the cycle used near the ground state approaches the same factor. We conjecture that Eq. (7) holds at every temperature. It holds in all our numerical tests, including exhaustive enumeration for $(d,d_M) = (2,3)$, $(2,4)$ and $(3,3)$, about 95 000 permutations for $(3,4)$ and $(4,4)$, and a search for violations over about 9 700 random resonant permutations (SM, Sec. S11). If it holds, the optimal quasistatic cost lies between $(L^*)^2/2(d_M-1)N$ and $(L^*)^2/2N$.

Discussion.—Thermodynamic length governs swap-based cooling [22], but for qudits it does not set the finite-resource limit. The leading-order cost is set instead by a Finsler geometry generated by resonant permutations, which improves on the Fisher geometry near

infinite temperature and near the ground state for every spectrum, and for the equally spaced qutrit and ququart at most intermediate temperatures as well (for the qutrit, everywhere outside $1.3 \lesssim \beta\omega \lesssim 2$). The cost still scales as $1/N$, but its prefactor is set by a different geometry, and the gain grows with the number of levels, approaching $d-1$ near the ground state, the largest gain that Theorem 2 allows at infinite temperature. Since the protocols only permute energy eigenstates, using machines whose gaps match system transitions, they suit platforms with equally spaced ladders, such as Zeeman sublevels of trapped-ion qudits [30] or bosonic modes. Permutations of energy eigenstates are also the elementary steps of heat-bath algorithmic cooling [9, 11], where work has mainly concerned the achievable polarization rather than the dissipation at finite N; the same analysis could be applied to its compression steps. In bosonic sub-bath cooling, exchanges of several excitations similarly improve on Gaussian operations [31]. To leading order around dissipationless collisions, Theorem 1 turns the remaining questions into classical ones: whether Eq. (7) holds at all temperatures for $d_M \geq 3$, what the norm is for $d \geq 4$ and for generic spectra at low temperature, and whether near-resonant protocols can exploit their offset from a resonance manifold.

AI tools were used in formalizing the mathematics and in drafting the paper. The authors take full responsibility for all content and have carefully checked every proof.

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SUPPLEMENTAL MATERIAL

S1. Proof of the one-step identity

Let $\rho'_{SM} = U(\rho \otimes \tau)U^\dagger$. Then $I(S:M)_{\rho'} + D(\rho'_M \,\|\, \tau) = S(\rho'_S) + S(\rho'_M) - S(\rho'_{SM}) - S(\rho'_M) - \operatorname{tr} \rho'_M \log \tau = S(\rho'_S) - S(\rho \otimes \tau) - \operatorname{tr} \rho'_M \log \tau$. On the other hand $D(\rho'_{SM} \,\|\, \rho \otimes \tau) = -S(\rho \otimes \tau) - \operatorname{tr} \rho'_S \log \rho - \operatorname{tr} \rho'_M \log \tau$ and $D(\rho'_S \,\|\, \rho) = -S(\rho'_S) - \operatorname{tr} \rho'_S \log \rho$. Subtracting gives Eq. (2). In the protocol, the machine M_n is uncorrelated with the system before step n, so its energy change is fixed by the bipartite state of S and M_n, which evolves from $\rho_{n-1} \otimes \tau_{M_n}$. Summing over n gives Eq. (1) with $\beta \Delta E_{M_n} = S(\rho_{n-1}) - S(\rho_n) + \Sigma_n$.

S2. Local expansion

For a permutation π of the configurations $c = (x, y)$ we have $r' = r \circ \pi^{-1}$ and, exactly, $\Sigma = D(r' \,\|\, r) - D(p' \,\|\, p)$. If π is resonant at q^*, i.e., $p_x q^*_y = p_{x'} q^*_{y'}$ whenever $\pi(x, y) = (x', y')$, then $r^* \circ \pi^{-1} = r^*$. Write $q = q^* + \delta q$ with $\sum_y \delta q_y = 0$. Then $\delta r_{\pi(c)} = p_{x(c)} \delta q_{y(c)} - p_{x(\pi c)} \delta q_{y(\pi c)}$ and $\delta p_x = \sum_y \delta r_{(x,y)}$, both linear in δq, and expanding both relative entropies to second order gives Eq. (3). Resonance is a linear condition on $\log q^*$: for each step $(x, y) \mapsto (x', y')$ of a nontrivial cycle, $\log q^*_y - \log q^*_{y'} = \log p_{x'} - \log p_x$. Among the $9! - 1$ non-identity qutrit permutations, these equations can be solved for 575 at a generic p, and 360 of these move the system. On the manifold $p_1^2 = p_0 p_2$ they can be solved for 809, of which 594 move the system. We enumerated all of them.

S3. Qubits

Of the 4! permutations of the configurations of two qubits, five admit a resonant machine for generic p. Three of them only relabel machine levels at a fixed system level and do not move the system. The other two are the swap $|0, 1\rangle \leftrightarrow |1, 0\rangle$, resonant at $q^* = p$, and the exchange $|0, 0\rangle \leftrightarrow |1, 1\rangle$, resonant at $q^* = (p_1, p_0)$. For both, Eq. (3) equals $\frac{1}{2}\|\delta p\|_F^2$, so the local norm is the Fisher norm, in agreement with the qubit optimality theorem of Ref. [22].

S4. Infinite temperature

At $p = q = (1/d, \ldots, 1/d)$ every permutation is resonant and $\delta r_{c'} = d^{-1}(\delta q_{y(\pi^{-1} c')} - \delta q_{y(c')})$. For a fixed π and a target direction g, minimizing Eq. (3) over δq subject to $\delta p \propto g$ is a quadratic program with a closed-form solution. For $d = 3$ and $g = (1, 0, -1)$ we evaluated all 9! permutations. Of these, 176 256 can move the system along g, and the minimum is exactly $1/2$ of the swap cost, attained for instance by π_{A} of Eq. (5). Section S10 extends this to mixtures and alternations of permutations, to all machine states and, via Theorem 1, to all collisions close to dissipationless ones. For $d \geq 4$ we minimized over π by simulated annealing. For the cooling direction of the equally spaced d-level system, $g_x \propto \bar{E} - E_x$, the best permutations found give ratios 0.567 ($d = 4$), 0.400 ($d = 5$) and 0.408 ($d = 6$); since the search is heuristic, these are only upper bounds. For transfers between the extreme levels, $d - 1$ parallel channels achieve the ratio $1/(d - 1)$.

S5. Protocols and numerics

We label configuration (x, y) by $c = 3x + y$ and list a permutation as $(\pi(0), \ldots, \pi(8))$. The protocols use $\pi_{\mathrm{A}} = (8, 6, 2, 3, 4, 5, 1, 7, 0)$ [Eq. (5)], $\pi_{\mathrm{B}} = (5, 3, 2, 8, 4, 0, 6, 7, 1)$, $\pi_{\mathrm{C}} = (5, 0, 2, 8, 3, 1, 6, 7, 4)$ and $\pi_{\mathrm{D}} = (8, 5, 2, 7, 0, 1, 6, 3, 4)$. Of these, π_{A} is resonant at every p, while π_{B}, π_{C} and π_{D} are resonant only on $p_1^2 = p_0 p_2$, where their resonance conditions become consistent. In Table I, the window $\beta\omega = 0.01 \to 0.05$ uses π_{A} in every step followed by one final swap, $0.1 \to 0.5$ alternates π_{B} and π_{A}, $3 \to 9$ alternates π_{C} and π_{D}, and $8 \to 16$ uses π_{C} in every step.

The N-scans of Fig. 1(b) use the same patterns, except that for $0.01 \to 0.05$ we use π_{A} in nine of every ten steps and a swap in the tenth, so that the middle level can also adjust. For this window we find $N\Sigma_N = 2.82 \times 10^{-4}$ at $N = 40$, 80 and 160, against $(L^*)^2/2 = 5.33 \times 10^{-4}$; the pattern of Table I gives 2.76, 2.81 and 2.93×10^{-4}. For $0.1 \to 0.5$ we find $N\Sigma_N = 0.0387$ at all three values of N, against 0.0507. For $3 \to 9$ we find $N\Sigma_N = 0.0718$, 0.0710 and 0.0706, against 0.0939, 0.0927 and 0.0921 for swaps, which tend to $(L^*)^2/2 = 0.0915$.

Machine populations were optimized with L-BFGS on the exact dynamics, using a penalty continuation for the final state. The swap optimum $\min \sum_n D(p_{n-1} \,\|\, p_n)$ is a convex problem in the intermediate states, which we solved with exact endpoints. We minimized over log-parametrized intermediate states, so that small populations are resolved, starting from uniformly spaced points on the Fisher–Rao geodesic, and re-evaluated the resulting path in 40-digit arithmetic. Where the populations are not small, the values agree with exponential-cone solutions to six or more digits, and an independent implementation with a second solver reproduces the table entries. Penalty-based or default-tolerance conic solutions can be inaccurate when populations are as small as 10^{-7}, and we do not use them.

The hull ratios of Fig. 1(a) are computed from a primal–dual pair of cone programs. On the Gibbs curve 809 permutations are resonant. For 764 of them the resonant machine state is unique, and for the other 45 it ranges over a line. The primal problem is a second-order cone program over all mechanisms with a unique machine state together with dense samples of the one-parameter families. It gives an attainable decomposition and hence an upper bound on the gauge. Its optimal dual direction u gives the lower bound $(u \cdot w)/H(u)$, where $H(u)$ is the exact support function of the hull, with each family maximized over its whole parameter line. The two bounds agree to within 2×10^{-7} at every plotted point ($\beta\omega \le 12$). For $\beta\omega \ge 6$ we use the variance form of Sec. S11, which remains well conditioned at small populations. At $\beta\omega = 14$ the bracket is only $[0.70773, 0.70810]$, and we do not plot this point.

S6. The low-temperature limit

Let $x = e^{-\beta\omega}$, so that $p \propto (1, x, x^2)$ lies on the resonance manifold. Write π_{C} as two 3-cycles, $C_1 : |0,0\rangle \to |1,2\rangle \to |0,1\rangle \to |0,0\rangle$ and $C_2 : |1,0\rangle \to |2,2\rangle \to |1,1\rangle \to |1,0\rangle$. It is resonant at $q^* \propto (x, x, 1)$, a machine whose two excited levels are degenerate. For $\delta q = (a, b, -a-b)$ the system moves by $\delta p_0 = -p_0 a - p_1(a+b)$, $\delta p_2 = p_1 a + p_2(a+b)$ and $\delta p_1 = -\delta p_0 - \delta p_2$. The configurations of C_1 change by $\delta r_{|1,2\rangle} = p_0 a + p_1(a+b)$, $\delta r_{|0,1\rangle} = -p_0 b - p_1(a+b)$ and $\delta r_{|0,0\rangle} = p_0(b-a)$. All three have weight $\simeq x$, while those of C_2 have weight $\simeq x^2$. Put $a = x\alpha$ and $b = x\gamma$. To leading order in x, Eq. (3) gives $\Sigma = \frac{x}{2}\left[\gamma^2 + (\gamma-\alpha)^2\right]$, while the swap cost of the same move is $\frac{x}{2}\alpha^2$. The minimum over γ is at $\gamma = \alpha/2$, i.e., at equal flow through the two channels $|0,0\rangle \to |1,2\rangle$ and $|1,2\rangle \to |0,1\rangle$, and equals $\frac{x}{4}\alpha^2$, half the swap cost. The level-2 component of the move differs from the Gibbs tangent only at relative order x. Correcting it (for instance with π_{D}) costs a fraction of the total that vanishes as $x \to 0$, so $\|\dot{p}\|_{\mathrm{eff}}/\|\dot{p}\|_F \to 1/\sqrt{2}$.

S7. The convex-hull norm and ququarts

If mechanisms k are alternated over many steps, the asymptotic cost of a move w is $\frac{1}{2N} g(w)^2$, where g is the norm whose unit ball is the convex hull of the unit sets of the mechanisms (the gauge of this hull). It is the value of the second-order cone program

$$g(w) = \min \sum_k \|u_k\| \quad \text{subject to} \quad \sum_k D\, M_k T_k\, u_k = D\, w,$$

where $\delta q_k = T_k u_k$ diagonalizes the k-th quadratic form, M_k maps δq_k to the system move, and $D = \mathrm{diag}(p)^{-1/2}$. For qutrits this reproduces the hull ratios of Fig. 1(a), e.g., 0.7341 at $\beta\omega = 0.1$, 0.8313 at $\beta\omega = 4$ and 0.7207 at $\beta\omega = 8$.

For the ququart we included every permutation that conserves $H_S + H_M$ for an integer machine spectrum e with $0 = e_0 \le e_1 \le e_2 \le e_3 \le 4\omega$. When a spectrum allowed more than 300 such permutations, we sampled 300 within its energy shells, for 8162 mechanisms in total. The resulting ratios are upper bounds on the attainable ratio: 0.845, 0.894, 0.973, 0.957 and 0.720 at $\beta\omega = 0.05$, 0.5, 1, 2 and 4. With configurations labeled $c = 4x + y$, the protocols of Table I are as follows. For $\beta\omega = 0.05 \to 0.25$ we alternate $\pi_{\mathrm{E}} = (0, 8, 1, 9, 4, 12, 5, 13, 2, 10, 3, 11, 6, 14, 7, 15)$, with machine spectrum $(0, 2, 2, 4)\omega$, and $\pi_{\mathrm{F}} = (0, 4, 9, 13, 1, 5, 6, 7, 8, 12, 10, 14, 2, 3, 11, 15)$, with machine spectrum $(0, 1, 3, 4)\omega$. For $\beta\omega = 3 \to 6$ we repeat the sequence $\pi_{\mathrm{G}}, \pi_{\mathrm{G}}, \pi_{\mathrm{H}}$, with $\pi_{\mathrm{G}} = (0, 2, 3, 4, 1, 6, 7, 8, 5, 10, 11, 12, 9, 15, 14, 13)$ [machine $(0, 1, 1, 1)\omega$] and $\pi_{\mathrm{H}} = (0, 4, 1, 8, 2, 3, 5, 12, 6, 10, 7, 14, 9, 13, 11, 15)$ [machine $(0, 1, 1, 2)\omega$]. Both protocols were re-evaluated in 40-digit arithmetic with explicit machine Hamiltonians. They reach the target to 10^{-11}, and the swap benchmarks are certified optimal by the convex program.

S8. Every dimension $d \ge 3$

Let p be uniform, so that every configuration has weight $1/d^2$, and take the linear machine perturbation $\delta q_y = \epsilon\,(y - \frac{d-1}{2})$. Under π_{ext} of Eq. (6), each of the $d-1$ transpositions $|0, y\rangle \leftrightarrow |d-1, y+1\rangle$ moves the mass $m =$

$(\delta q_y - \delta q_{y+1})/d = -\epsilon/d$. The flow is therefore the same in all channels, and the system moves by $\delta p = M(e_0 - e_{d-1})$ with $M = (d-1)\epsilon/d$. Only the $2(d-1)$ exchanged configurations change, so Eq. (3) gives

$$\Sigma = \tfrac{1}{2}\Big[d^2 \cdot 2(d-1)\frac{\epsilon^2}{d^2} - d \cdot 2M^2\Big] = \frac{(d-1)\epsilon^2}{d},$$

whereas the swap cost of the same move is $\frac{1}{2}\sum_x \delta p_x^2/p_x = dM^2 = (d-1)^2\epsilon^2/d$. The ratio is $1/(d-1)$. The resonance conditions $p_0 q_y = p_{d-1} q_{y+1}$ can be solved for every p with a geometric machine, so $\pi_{\rm ext}$ can be used at every step without leaving its resonance manifold.

Now combine $\pi_{\rm ext}$ with swaps. In Fisher-normalized coordinates, let the cooling direction w and the extreme transfer $u = e_0 - e_{d-1}$ be unit vectors at angle θ_d. The attainable cost per unit length is

$$g(w) = \min_{\epsilon \geq 0} \big[a\,\epsilon + |w - \epsilon u|\big], \qquad a = (d-1)^{-1/2} = \cos\varphi_d.$$

If $a < \cos\theta_d$ the minimum is $\cos(\theta_d - \varphi_d) < 1$. At uniform p the Fisher metric is d times the Euclidean one and $w \propto (\bar{E} - E_x)_x$, which gives $\cos\theta_d = (d-1)/\sqrt{2\sum_x (x-\bar{E})^2} = \sqrt{6(d-1)/(d(d+1))}$; the condition $a < \cos\theta_d$ is equivalent to $5d^2 - 13d + 6 > 0$. Both mechanisms are resonant at every p, and the local quadratic forms depend continuously on p. For cooling windows close enough to infinite temperature, the leading coefficient of protocols that alternate swaps and $\pi_{\rm ext}$ is therefore below the thermodynamic-length value $(L^*)^2/2$. For $d = 3$ the construction gives $1/\sqrt{2}$, the exhaustive optimum of Sec. S4.

Arbitrary spectra. Let $E_0 < \cdots < E_{d-1}$ be arbitrary, with system levels labeled by energy. The resonance conditions of $\pi_{\rm ext}$ involve only p_0/p_{d-1}. They are solved by a machine whose levels are equally spaced by $E_{d-1} - E_0$, so $\pi_{\rm ext}$ conserves $H_S + H_M$ and is resonant at every p. At uniform p the computation above does not involve the spectrum, so the cost ratio of the extreme transfer is again $1/(d-1)$. The cooling direction at uniform p is $w \propto (\bar{E} - E_x)_x$, and $\cos\theta = (E_{d-1} - E_0)/\sqrt{2\sum_x (E_x - \bar{E})^2}$. Since d numbers in an interval of length R satisfy $\sum_x (E_x - \bar{E})^2 \leq dR^2/4$, we get $\cos\theta \geq \sqrt{2/d}$, which exceeds $a = (d-1)^{-1/2}$ for all $d \geq 3$ because $2(d-1) > d$. Hence swaps are beaten near infinite temperature for every spectrum. For the qutrit, $\cos\theta = 1$ for equally spaced levels, giving cost ratio $1/2$, and $\cos\theta \geq \sqrt{3}/2$ in general, giving a cost ratio of at most $\cos^2 15^\circ \approx 0.93$. The exact protocols quoted in the main text use $\pi_{\rm ext}$ with machine populations $q_y \propto (p_0/p_{d-1})^y$ at resonance.

S9. Near the ground state, every d

Let $x = p_1/p_0 \to 0$, with higher populations of higher order, and consider the d-cycle

$$C: \ |0, e_0\rangle \to |1, g\rangle \to |0, e_1\rangle \to \cdots \to |0, e_{d-2}\rangle \to |0, e_0\rangle.$$

Take a machine with $q_{e_j} = q_e$ and $q_e/q_g = p_1/p_0$. Since $q_g \simeq 1$, all d configurations of C then have the same weight $w \simeq x$.

Along a cycle, a perturbation of the weights changes configuration c_{i+1} by $a_{i+1} = \delta r(c_i) - \delta r(c_{i+1})$, with $\sum_i a_i = 0$. The configurations of C use d different machine levels, so δq can realize any such vector. The system moves by $\delta p_1 = a_{|1,g\rangle} =: M$ and $\delta p_0 = -M$.

In Eq. (3), the term $a_{|1,g\rangle}^2/2w = M^2/2x$ is canceled at leading order by $\delta p_1^2/2p_1 = M^2/2x$, and $\delta p_0^2/2p_0 = O(M^2)$ is negligible. What remains is $\sum_j a_{|0,e_j\rangle}^2/2w$ with $\sum_j a_{|0,e_j\rangle} = -M$. Equal shares $-M/(d-1)$ minimize it, giving $\Sigma = M^2/[2x(d-1)]$. The swap cost of the same move is $M^2/2x$ to leading order, so the ratio is $1/(d-1)$. Numerically, at $\beta\omega = 14$ the ratio is 0.5000, 0.3333, 0.2500 and 0.2000 for $d = 3, \ldots, 6$.

The components of the cooling direction on levels ≥ 2 are smaller in Fisher norm by a factor of order $e^{-\beta(E_2 - E_1)/2}$. Correcting them with occasional swaps costs a fraction of the total that vanishes as $\beta \to \infty$. For $d = 3$ the cycle is the first 3-cycle of $\pi_{\rm C}$.

The protocols of Table I and Fig. 2 for $\beta\omega = 8 \to 16$ are built as follows. For $d \geq 4$ they use the d-cycle in seven of every eight steps and a swap in the eighth, which adjusts the higher levels. The machine of each cycle step starts at the resonant state $q_e/q_g = p_1/p_0$, after which all machine populations are optimized. For $d = 3$ the table uses $\pi_{\rm C}$ in every step; the cycle protocol gives a ratio of 0.545 there. Against the swap optimum, which is 1.6288×10^{-5} for every d in this window, the ratios are 0.503, 0.372, 0.283, 0.229, 0.193 and 0.167 for $d = 3, \ldots, 8$.

S10. Coherent collisions: proof of Theorem 1

Throughout, the system and machine have arbitrary finite dimensions d and d_M, and all states are full rank. For a state σ, the Kubo–Mori norm is $\|A\|_\sigma^2 = \int_0^\infty \mathrm{tr}[A(\sigma+t)^{-1}A(\sigma+t)^{-1}]\,dt$. For diagonal $\sigma = \mathrm{diag}(s)$ it equals $\sum_{ij}|A_{ij}|^2 c(s_i,s_j)$, with $c(a,b) = (\log a - \log b)/(a-b)$ and $c(a,a) = 1/a$. It is the Hessian of the relative entropy, $D(\sigma + A \,\|\, \sigma + B) = \frac{1}{2}\|A-B\|_\sigma^2 + O(\|A\|^3 + \|B\|^3)$ for traceless Hermitian A, B, and it is monotone under completely positive trace-preserving maps [28].

Dissipationless collisions. By Eq. (2) and $S(\rho'_{SM}) = S(\rho\otimes\tau)$, we have $\Sigma = D(\rho'_{SM} \,\|\, \rho'_S \otimes \tau)$, so $\Sigma = 0$ if and only if $\rho'_{SM} = \rho'_S \otimes \tau$. In that case $\rho\otimes\tau$ and $\rho'_S\otimes\tau$ have the same spectrum. Comparing $\mathrm{tr}(\cdot)^k$ for all k gives $\mathrm{tr}\,\rho'^k_S = \mathrm{tr}\,\rho^k$, so $\rho'_S = V\rho V^\dagger$ for a unitary V, and $U_0 := (V^\dagger\otimes 1)U$ commutes with $\rho\otimes\tau$. Conversely, every such U is dissipationless. A system unitary V changes neither Σ nor the spectrum of the system, so we set $V = 1$.

Second order. Let $\rho^* = \mathrm{diag}(p^*)$ and $\tau^* = \mathrm{diag}(q^*)$, put $s^* = \rho^*\otimes\tau^*$, and let U_0 commute with s^*. To include near-resonant collisions, we allow the incoming state to differ from ρ^*, $\rho = \rho^* + \epsilon R$ with R diagonal, where either $R = 0$ (the case of the main text, any p^*) or p^* is non-degenerate. Let $\tau = \tau^* + \epsilon\Theta$ with Θ diagonal and traceless, and $U = U_0 e^{i\epsilon K}$. Then $\rho\otimes\tau = s^* + \epsilon D + O(\epsilon^2)$ with $D = R\otimes\tau^* + \rho^*\otimes\Theta$, and

$$U(\rho\otimes\tau)U^\dagger - \rho\otimes\tau = \epsilon X + O(\epsilon^2), \qquad X = \underbrace{U_0 D U_0^\dagger - D}_{X_\parallel} + \underbrace{U_0\, i[K,s^*]\, U_0^\dagger}_{X_\perp}.$$

Expanding both relative entropies of Eq. (2) around s^* and ρ^* gives $\Sigma = \epsilon^2 Q + O(\epsilon^3)$ with $Q = \frac{1}{2}\|X\|_{s^*}^2 - \frac{1}{2}\|\mathrm{tr}_M X\|_{\rho^*}^2$, which is Eq. (4) for $R = 0$.

Orthogonal splitting. Group the product basis $|x,y\rangle$ into the eigenspaces (blocks) of s^*. The matrix $i[K,s^*]$ has entries $iK_{ij}(s^*_j - s^*_i)$, which vanish inside blocks. Since U_0 is block diagonal, $X_\parallel$ lives inside the blocks and $X_\perp$ between them. For diagonal s^* the Kubo–Mori norm is a weighted sum over entries, so $\|X\|_{s^*}^2 = \|X_\parallel\|_{s^*}^2 + \|X_\perp\|_{s^*}^2$. An entry of $X_\parallel$ joining $|x,y\rangle$ and $|x',y\rangle$ requires $p^*_x q^*_y = p^*_{x'} q^*_y$, i.e., $p^*_x = p^*_{x'}$. Hence $\mathrm{tr}_M X_\parallel$ lives inside the eigenspaces of ρ^*. Conversely, $\mathrm{tr}_M X_\perp$ has no entries there, because $|x,y\rangle$ and $|x',y\rangle$ with $p^*_x = p^*_{x'}$ lie in one block. So $Q = Q_\parallel + Q_\perp$, where $Q_\bullet = \frac{1}{2}\|X_\bullet\|_{s^*}^2 - \frac{1}{2}\|\mathrm{tr}_M X_\bullet\|_{\rho^*}^2$.

(i) Coherent part. $Q_\perp \geq 0$, by monotonicity of the Kubo–Mori metric under the partial trace, which maps s^* to ρ^*.

(ii) Block part. Conjugating by $W\otimes 1$, with W unitary inside the eigenspaces of ρ^* (this commutes with s^* and with D, since $R = 0$ whenever p^* is degenerate), we may assume that $\mathrm{tr}_M X_\parallel$ is diagonal. Inside a block with eigenvalue r, $\|\cdot\|_{s^*}^2 = r^{-1}\|\cdot\|_{\mathrm{HS}}^2$. For diagonal D, $\|U_0 D U_0^\dagger - D\|_{\mathrm{HS}}^2 = 2\,\mathrm{tr}\,D^2 - 2\sum_{ij} B_{ij} D_i D_j$, where $B_{ij} = |(U_0)_{ij}|^2$ is doubly stochastic and block diagonal. By Birkhoff's theorem [29], $B = \sum_k \lambda_k P_k$, a convex combination of block-diagonal permutation matrices. Each P_k commutes with s^*, and $\|X_\parallel\|_{s^*}^2 = \sum_k \lambda_k \|P_k D P_k^{\mathrm{T}} - D\|_{s^*}^2$. The diagonal of $U_0 D U_0^\dagger$ is $B\,\mathrm{diag}(D)$, so $\mathrm{tr}_M X_\parallel = \mathrm{diag}(\bar\delta)$ with $\bar\delta = \sum_k \lambda_k \delta_k$ and $\delta_k = \mathrm{tr}_M(P_k D P_k^{\mathrm{T}} - D)$. By convexity of $\delta \mapsto \sum_x \delta_x^2/p^*_x$,

$$Q_\parallel = \tfrac{1}{2}\sum_k \lambda_k \|P_k D P_k^{\mathrm{T}} - D\|_{s^*}^2 - \tfrac{1}{2}\|\bar\delta\|_{\rho^*}^2 \;\geq\; \sum_k \lambda_k Q_k, \qquad Q_k = \tfrac{1}{2}\|P_k D P_k^{\mathrm{T}} - D\|_{s^*}^2 - \tfrac{1}{2}\|\delta_k\|_{\rho^*}^2.$$

For $R = 0$, Q_k is the cost (3) of P_k with machine perturbation Θ. Equality holds if and only if all δ_k coincide.

(iii) Spectral move. To first order, the eigenvalues of $\rho'_S = \rho + \epsilon\,\mathrm{tr}_M X + O(\epsilon^2)$ are those of $\rho^* + \epsilon(R + \mathrm{tr}_M X)$ compressed to the eigenspaces of ρ^*. Since $\mathrm{tr}_M X_\perp$ does not contribute, the spectrum moves by $\epsilon\bar\delta$. Together, (i)–(iii) prove Theorem 1, including for near-resonant collisions.

Consequences. For $R = 0$, each P_k with perturbation Θ is one of the mechanisms whose unit sets define $\|\cdot\|_{\mathrm{eff}}$, so $Q_k \geq \frac{1}{2}\|\delta_k\|_{\mathrm{eff}}^2$ and

$$Q \;\geq\; \sum_k \lambda_k Q_k \;\geq\; \tfrac{1}{2}\sum_k \lambda_k \|\delta_k\|_{\mathrm{eff}}^2 \;\geq\; \tfrac{1}{2}\Big(\sum_k \lambda_k \|\delta_k\|_{\mathrm{eff}}\Big)^2 \;\geq\; \tfrac{1}{2}\|\bar\delta\|_{\mathrm{eff}}^2.$$

For the quasistatic regime, assume that every collision is an $O(1/N)$ perturbation of a collision that is dissipationless for its incoming state, that system and machine populations are bounded away from zero, and that K and Θ are uniformly bounded, so that the remainders are uniform. Summing over steps and using Cauchy–Schwarz gives $\sum_n \Sigma_n \geq \frac{1}{2}\sum_n \|\delta p_n\|_{\mathrm{eff}}^2 - O(N^{-2}) \geq \ell^2/2N - O(N^{-2})$, with $\ell = \sum_n \|\delta p_n\|_{\mathrm{eff}}$. Conversely, suppose that permutations P_k remain resonant along a path, with resonant machines depending continuously on p. Using each P_k in a fraction λ_k of the steps then reproduces the first-order moves and the cost $\sum_k \lambda_k Q_k$, without the Jensen gap of (ii). Taking the P_k, weights and moves from an optimal decomposition of the hull gauge attains $\frac{1}{2}\|\delta p\|_{\mathrm{eff}}^2$ per step on average. The

permutations P_k act on the product basis after the rotation W of (ii), i.e., on an eigenbasis of ρ^* rotated within its degenerate eigenspaces.

Numerical checks. We drew random dissipationless collisions for $(d, d_M) \in \{(3,3),(4,4),(3,4),(4,3),(3,2)\}$, with resonances of swap, extreme-level ($\pi_{\rm ext}$-type), partial and uniform-machine type, including degenerate system spectra, and with random block unitaries U_0 and random K and Θ, at least 150 draws per dimension pair. In double precision, the identity $Q_\parallel = \sum_k \lambda_k Q_k + \frac{1}{2}\big(\sum_k \lambda_k \|\delta_k\|^2_{\rho^*} - \|\bar\delta\|^2_{\rho^*}\big)$ held to relative accuracy 3×10^{-14}, $Q_\perp$ was nonnegative in every draw, and the Birkhoff move $\sum_k \lambda_k \delta_k$ agreed with the first-order move computed from X to 3×10^{-14}. We also compared Eq. (4) with the exact dissipation in 60-digit arithmetic, using exactly resonant machines and Richardson extrapolation in ϵ, for 18 draws with $(d, d_M) = (4,4),(3,4),(4,3),(3,3)$ that cover swap, extreme-level and partial resonances and degenerate system spectra. The two agreed to at least 14 significant digits, with machine populations as small as 10^{-16}. For the near-resonant case $R \neq 0$, nine draws in 50-digit arithmetic agreed with the formula to 10^{-21}, satisfied $Q \geq \sum_k \lambda_k Q_k$, and had first-order moves equal to the Birkhoff mixture.

Near-resonant offsets. The qutrit Gibbs curve lies on the resonance manifold $p_1^2 = p_0 p_2$, and a protocol may stay $O(1/N)$ away from it. A permutation that is resonant only on the manifold then acts with an offset R along its normal. The offset enters $D = R \otimes \tau^* + \rho^* \otimes \Theta$ and can unlock moves that are blocked on the manifold. To bound the possible gain, we let every mechanism choose its own offset, which is an optimistic relaxation. The hull ratio of Fig. 1(a) then decreases as follows:

$\beta\omega$	0.02	0.1	0.5	1.6	4	8
exactly resonant	0.7120	0.7341	0.8866	1	0.8313	0.7207
relaxed offsets	0.7110	0.7297	0.8778	1	0.8014	0.7099

Whether actual protocols can realize any of this is open. In all cases the relaxed values stay above the conjectured bound $1/\sqrt{2}$.

Infinite temperature: a dual certificate. Let $p = (\frac{1}{3},\frac{1}{3},\frac{1}{3})$ and $g = (1,0,-1)$, the cooling direction of the equally spaced qutrit. Let h be the largest value of $\langle g,\delta\rangle_F/\sqrt{2Q}$ over all mechanisms and perturbations, where $\langle g,\delta\rangle_F = \sum_x g_x\delta_x/p_x$. The linear functional $\langle g,\cdot\rangle_F/h$ is then at most 1 on the hull, so $\|g\|_{\rm eff} \geq \|g\|_F^2/h$. For a single mechanism, $h^2 = b^{\rm T}A^+b$, where $2Q = \Theta^{\rm T}A\,\Theta$ and $\langle g,\delta\rangle_F = b^{\rm T}\Theta$; it is infinite if A has a null vector with $b^{\rm T}\Theta \neq 0$. Three cases occur. (i) For a uniform machine, $q^* = p$, all 9! permutations are resonant. We evaluated h^2 for each of them, in floating point and, independently, in exact rational arithmetic. None is infinite. The largest value is exactly $2\|g\|_F^2 = 12$, attained by 1296 permutations including $\pi_{\rm A}$, and the next value is $15/2 = 1.25\|g\|_F^2$. (ii) If exactly two machine populations are equal, say $q_0^* = q_1^* = a < \frac{1}{2}$ after relabeling, only exchanges between machine levels 0 and 1 move the system. Each changes a configuration by $\pm t$, with $t = (\theta_1 - \theta_0)/3$. The move is $\delta = t\,v$ with $v_x \in \{-1,0,1\}$, and $2Q = (6k/a - 3|v|^2)t^2$, where $k \geq 1$ is the number of exchanges in each direction. Since $(v_0 - v_2)^2 \leq 4$ and $|v|^2 \leq 2 \leq 2k$, we get $h^2 \leq 36/(6/a - 6) = 6a/(1-a) < 6 = \|g\|_F^2$. (iii) If the three machine populations are distinct, no resonant permutation moves the system. Therefore $\|g\|_{\rm eff} \geq \|g\|_F/\sqrt{2}$, and $\pi_{\rm A}$ attains this value. With qutrit machines at the maximally mixed state, half the swap cost is thus the optimum over the hull, and by Theorem 1 also over all collisions close to dissipationless ones. With d_M-level machines the optimum is at least $1/(d_M - 1)$ of the swap cost (Theorem 2); for $d_M \geq 4$, three or more $\pi_{\rm ext}$-type channels can go below $\frac{1}{2}$.

S11. Maximal gain: proof of Theorem 2

Let P be a permutation of the configurations that commutes with $s^* = \rho^* \otimes \tau^*$, and let Θ be diagonal and traceless. Write $\kappa(c)$ for the configuration whose population flows into c, so that $\kappa = P^{-1}$ and $r^*_{\kappa(c)} = r^*_c$. With $D = \rho^* \otimes \Theta$ we have $X_c = D_{\kappa(c)} - D_c$. For $c = (x,y)$ and $\kappa(c) = (x',y')$, resonance gives $p^*_{x'}q^*_{y'} = p^*_x q^*_y$, so

$$X_{(x,y)} = r^*_{(x,y)}\big[\psi(y') - \psi(y)\big], \qquad \psi = \Theta/q^*, \qquad \textstyle\sum_y q^*_y\psi(y) = \operatorname{tr}\Theta = 0.$$

For each system level x let $\sigma_x(y)$ be the machine level of $\kappa(x,y)$, and put $\Delta_x(y) = \psi(\sigma_x(y)) - \psi(y)$. Then $\delta_x = p^*_x\,\mathbb{E}_{q^*}\Delta_x$. With expectations over $y \sim q^*$,

$$2Q = \sum_x p^*_x\Big[\mathbb{E}(\Delta_x^2) - (\mathbb{E}\Delta_x)^2\Big], \qquad \|\delta\|_F^2 = \sum_x p^*_x(\mathbb{E}\Delta_x)^2.$$

Hence $\|\delta\|_F^2 \leq (d_M - 1)\,2Q$ is equivalent to

$$d_M \sum_x p^*_x(\mathbb{E}\Delta_x)^2 \;\leq\; (d_M - 1)\sum_x p^*_x\,\mathbb{E}(\Delta_x^2). \tag{S11.1}$$

(a) Maximally mixed system. Here $p_x^* = 1/d$, and resonance forces $q^*_{\sigma_x(y)} = q^*_y$. So σ_x maps each class $\mathcal{K}$ of machine levels with a common population $q_{\mathcal{K}}$ into itself. Choose a cycle $C_{\mathcal{K}} \subseteq \mathcal{K}$ of σ_x in each class. Along a cycle the differences Δ_x telescope to zero, and q^* is constant on it, so the cycles do not contribute to $\mathbb{E}\Delta_x$. By Cauchy–Schwarz over the remaining levels S_x,

$$(\mathbb{E}\Delta_x)^2 \le \Big(\sum_{y\in S_x} q_y^*\Big)\,\mathbb{E}(\Delta_x^2) \le \Big(1-\sum_{\mathcal{K}} q_{\mathcal{K}}\Big)\,\mathbb{E}(\Delta_x^2) \le \Big(1-\frac{1}{d_M}\Big)\,\mathbb{E}(\Delta_x^2),$$

since $1 = \sum_{\mathcal{K}} |\mathcal{K}|\, q_{\mathcal{K}} \le d_M \sum_{\mathcal{K}} q_{\mathcal{K}}$. Summing over x gives (S11.1).

(b) Qubit machines, any system state. For $d_M = 2$, write $\psi = a\,(q_1^*, -q_0^*)$. Each σ_x is one of the four maps of $\{0,1\}$ into itself:

map	$(\mathbb{E}\Delta_x)^2/a^2$	$\mathbb{E}(\Delta_x^2)/a^2$
identity	0	0
exchange	$(q_1^* - q_0^*)^2$	1
constant 0	q_1^{*2}	q_1^*
constant 1	q_0^{*2}	q_0^*

Let P_{id}, P_{ex}, P_0 and P_1 be the total p^*-weights of the four types. The inflowing configurations $\kappa(c)$ are again distributed as r^*, so the machine level $\sigma_X(Y)$ of $(X,Y) \sim r^*$ has law q^*:

$$P_{\mathrm{id}} q_1^* + P_{\mathrm{ex}} q_0^* + P_1 = q_1^*, \qquad \text{equivalently} \qquad P_{\mathrm{ex}}(q_0^* - q_1^*) + P_1 q_0^* = P_0 q_1^*.$$

Substituting this gives

$$\sum_x p_x^* (\mathbb{E}\Delta_x)^2 = a^2\big[P_{\mathrm{ex}} q_0^*(q_0^* - q_1^*) + P_1 q_0^*\big], \qquad \sum_x p_x^* \mathbb{E}(\Delta_x^2) = 2a^2(P_{\mathrm{ex}} + P_1) q_0^*.$$

Then (S11.1) reads $P_{\mathrm{ex}} q_0^*(q_0^* - q_1^*) \le P_{\mathrm{ex}} q_0^*$, which always holds.

(c) From permutations to collisions. By Theorem 1 (with $R = 0$, for any p^*) and convexity, $Q \ge \sum_k \lambda_k Q_k \ge \frac{1}{d_M - 1}\,\frac{1}{2}\sum_k \lambda_k \|\delta_k\|_F^2 \ge \frac{1}{d_M-1}\,\frac{1}{2}\|\bar{\delta}\|_F^2$.

Attainment. For any d and d_M, take the maximally mixed system, a uniform machine and the exchange $|0, y\rangle \leftrightarrow |d-1, y+1\rangle$ for $y = 0, \dots, d_M - 2$, with $\theta_y = \epsilon\,(y - \frac{d_M - 1}{2})$. Each exchanged configuration changes by $\pm\epsilon/d$, and the system moves by $\delta_0 = -\delta_{d-1} = (d_M - 1)\epsilon/d$. Then $\sum_c^{\prime} X_c^2/r_c = 2(d_M - 1) d_M \epsilon^2/d$ and $\|\delta\|_F^2 = 2(d_M - 1)^2\epsilon^2/d$, so $\|\delta\|_F^2/2Q = d_M - 1$, which is equality in Eq. (7). For $d_M = d$ this exchange is π_{ext} (Sec. S8).

No free moves. For an exactly resonant permutation, $2Q = 0$ forces $\delta = 0$, so $\max_\Theta \|\delta\|_F^2/2Q$ is finite at every state. Indeed, $2Q = 0$ means that each Δ_x is a constant c_x, i.e., $\psi(\sigma_x(y)) = \psi(y) + c_x$. Take $X \sim p^*$ independent of $Y \sim q^*$. The marginal condition $\sigma_X(Y) \sim q^*$ then gives $\mathbb{E}\,\psi(Y)^2 = \mathbb{E}\,\psi(\sigma_X(Y))^2 = \mathbb{E}\,[\psi(Y) + c_X]^2 = \mathbb{E}\,\psi(Y)^2 + \mathbb{E}\,c_X^2$, using $\mathbb{E}\,\psi(Y) = 0$. Hence all $c_x = 0$ and $\delta = 0$.

(d) Numerical evidence at general temperature. We tested Eq. (S11.1) by running over all permutations of the dd_M configurations. At each test state we solved the resonance equations, sampled the resonance families, and computed $\max_\Theta \|\delta\|_F^2/2Q$. The test states were the maximally mixed state, Gibbs states of the equally spaced system at several temperatures, Gibbs states of random spectra, and random states with all configuration weights above 10^{-9}. (At the maximally mixed state the enumeration is incomplete, but that case is covered by (a).) For $(d, d_M) = (2,3)$ and $(2,4)$ the value $d_M - 1$ is attained at the maximally mixed state, while elsewhere the maxima stay below it and approach it toward the ground state (1.985 and 2.942 at $\beta\omega = 6$). For $(3,3)$ the maxima stay below 2, the largest being 1.99996 at $\beta\omega = 6$. The cases $(3,2)$ and $(4,2)$, covered by (b), serve as checks and give at most 1. One double-precision run at $\beta\omega = 12$ returned an infinite value. This is an artifact of configuration weights below 10^{-10}, since by the lemma above infinite values cannot occur.

We repeated the $(3,3)$ test along the Gibbs curve in 60-digit arithmetic, with exactly resonant machines, over all 594 resonant permutations that move the system. No infinite value occurs, and the largest value lies below the bound 2 by

$\beta\omega$	0.05	6	12	16	24	40
$2 - \max\lambda$	1.2×10^{-3}	3.7×10^{-5}	2.3×10^{-10}	7.6×10^{-14}	8.6×10^{-21}	1.1×10^{-34}

The bound is thus approached from below at both ends of the Gibbs curve. For $(d, d_M) = (3,4)$ and $(4,4)$ exhaustive enumeration is out of reach. There we took all permutations that conserve $H_S + H_M$ for integer machine spectra up

to 4ω, about 5 300 and 9 500 per temperature, at $\beta\omega = 0.05$, 0.5, 2 and 5. The largest values are 2.9989 and 2.99999, below the bound 3, and the cases flagged in double precision were confirmed in 50-digit arithmetic. Repeating this for random non-degenerate integer system spectra and temperatures, with about 35 500 mechanisms for $d = 3, 4$, gives a largest value of 2.995.

The most demanding test is a direct search for violations. For a fixed permutation, the resonance conditions are linear in $(\log p, \log q^*)$. For each of $(d, d_M) = (2,4), (2,5), (3,3), (3,4), (4,3), (4,4)$ we drew 3000 random sparse permutations, products of up to four short cycles. For each of the 9 687 that are resonant on a nontrivial set, we searched (Nelder–Mead from three starting points) for the largest $\max_\Theta \|\delta\|_F^2/2Q$ over all system and machine states at which the permutation is resonant. The maxima are 3.00000000, 2.99999998, 2.00000000, 2.99999997, 2.00000000 and 2.9995, against bounds 3, 4, 2, 3, 2 and 3. The bound is never exceeded, and it is reached for $(2,4)$, $(3,3)$, $(3,4)$ and $(4,3)$; for $(2,5)$ the sample missed the four-channel exchange that attains 4.

The marginal condition alone does not suffice. Part (b) uses only the marginal condition $\sigma_X(Y) \sim q^*$, so one may ask whether arbitrary maps σ_x subject to this condition obey (S11.1). For fixed q^* and ψ, the worst family of maps maximizes $\sum_x p_x(\mathbb{E}\Delta_x)^2 / \sum_x p_x \mathbb{E}(\Delta_x^2)$, a linear-fractional program over distributions on all $d_M^{d_M}$ maps. We maximized it over q^* and ψ and then recomputed the optimal vertices in 50-digit arithmetic. For $d_M = 3$ the ratio approaches $\frac{2}{3}$ only from below; the largest value, at the boundary of the simplex, is $\frac{2}{3} - 2.9 \times 10^{-10}$. For $d_M = 4$, however, it exceeds $\frac{3}{4}$. For example, $q^* \approx (0.4386, 0.0878, 0.1782, 0.2954)$ with four maps of weights 0.73, 0.17, 0.04 and 0.06 gives 0.7553, with the marginal condition satisfied to 10^{-50}. These families violate weight preservation, so they do not come from resonant permutations. For $d_M = 2$ the LP-dual certificate $g = \frac{1}{2}\psi^2$, which satisfies $(\mathbb{E}\,\psi \circ \sigma)^2 \le \mathbb{E}\big[\frac{1}{2}(\psi \circ \sigma - \psi)^2 + g \circ \sigma - g\big]$ for every map σ, gives a second proof of (b). For $d_M \ge 4$, a proof of the general case must therefore use the full resonance structure (bijectivity and weight preservation), not only the marginal condition. Random sampling misses these counterexamples, which shows how special they are.

S12. Code and data

All numerical results in the Letter and in this Supplemental Material were obtained with Python scripts, which are available together with their output logs and the protocol data at https://github.com/anshM123/Resonant-Collision-Cooling. The README file there lists the script behind each statement.